Muhammad Andrata Zharfan Mustika

AI Engineer | Data Scientist | Data Analyst
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SUMMARY

Informatics Engineering graduate (S1) with a strong interest in AI and data, experienced in building LLM and RAG systems, machine learning for text and image data, and data analysis from preprocessing through evaluation. Comfortable leading teams, translating business requirements into technical solutions, and presenting results to technical and non-technical audiences. Seeking roles in AI engineering, data science, or data analysis.

EDUCATION

Universitas Komputer Indonesia (UNIKOM), Bandung *2022 - 2026*
Bachelor of Computer Science (S.Kom.) in Informatics Engineering
GPA: 3.75 / 4.00

WORK & ORGANIZATIONAL EXPERIENCE

Fullstack Web Developer Intern *Aug 2025 - Nov 2025*
Rama Group (Rama FM & Paramuda Radio)

- Led development of the Rama FM and Paramuda Radio web platforms, covering system architecture, UI/UX improvements, and content management with Laravel, Inertia, and React.
- Served as the main technical liaison for stakeholders: coordinated the development team, managed schedules, and translated business requirements into technical solutions.
- Collaborated with the marketing division to position the website as a promotion and branding channel, and took part in client meetings.

General Secretary *2024 - 2025*
Himpunan Mahasiswa Teknik Informatika (HMIF) UNIKOM

- Managed organizational administration: correspondence, meeting minutes, and activity documentation.
- Archived important documents (digital and physical) and supported the chair in coordinating work programs, internal policies, and meetings.

Member, Faction 1 FTIK (MPM) & Commission 5 (DPM) *2023 - 2024*
MPM & DPM UNIKOM

- Channeled student aspirations for the Faculty of Engineering and Computer Science as a Member of Faction 1 FTIK in the Student Consultative Assembly (Majelis Permusyawaratan Mahasiswa, MPM).
- As a Member of Commission 5 (Creative Economy, Research, and IT) of the Student Representative Council (Dewan Perwakilan Mahasiswa, DPM), oversaw the Tekraf Ministry (Creative Economy and Communication & Informatics), drafted policy, and managed DPM's social media and communications.

PROJECTS

Implementation of Retrieval-Augmented Generation (RAG) with a Qwen 2.5 7B Instruct Generator in a Stunting Question-Answering System (Undergraduate Thesis) *2025 - 2026*
Python · LLM · LangChain · RAG · RAGAS · FAISS · Gradio

- Engineered an end-to-end Indonesian RAG pipeline to answer stunting & public health queries from 60 official policy documents and journals, indexing 12,558 vectorized chunks using multilingual-e5-base and FAISS.
- Curated an 80-question Gold Standard dataset across 6 query categories, validated with a domain expert (nutritionist) using a Human-in-the-Loop (HITL) protocol.
- Evaluated the system with RAGAS (0.95 Retrieval MRR, 0.96 Context Precision, 0.90 Faithfulness) and compared it with a No-RAG baseline on the same 80 questions: Answer Relevancy rose from 0.73 to 0.92 and Answer Correctness from 0.53 to 0.73; numerical questions were the main weakness (0.69 Faithfulness).

Early Detection of Heart Disease *2026*

Python · XGBoost · SHAP · MLJAR AutoML

- Built an early-detection classification model for heart disease on 908 patient records (11 features) with SHAP-based feature selection, then compared Decision Tree, Random Forest, Logistic Regression, and AutoML; Decision Tree (max_depth=3) performed best with 84.1% accuracy and 87.5% recall on the test set.

Child Stunting Status Classification from Basic Anthropometric Data *2026*

Python · Scikit-learn · Pandas · Seaborn

- Built a four-class classification of stunting status for children aged 0-24 months from four basic anthropometric variables on 92,692 records, comparing Logistic Regression (macro-F1 0.58) and Random Forest (macro-F1 of about 1.00), and critically assessing the near-perfect score because labels were derived from z-score rules on a synthetic dataset.

Sentiment Analysis of the Film Jumbo on Twitter (X) *2025*

Python · Scikit-learn · Pandas · Sastrawi · NLTK · imbalanced-learn

- Built a three-class sentiment classification pipeline on 5,302 Indonesian-language tweets (CRISP-DM), covering preprocessing (slang normalization, Sastrawi stemming, negation handling), TF-IDF unigram-bigram features, and Logistic Regression.
- Addressed class imbalance (negative class only 5.4% of data) with class weights, SMOTE, RandomOverSampler, and 5-fold GridSearchCV: negative-class recall rose from 0.12 to 0.58, with 69% accuracy and macro-F1 of 0.63 on the test set.

Face Image Classification with CNN (ResNet50) *2025*

Python · PyTorch · Convolutional Neural Networks · OpenCV · Scikit-learn

- Built a three-class face image classification pipeline on 86,744 images, covering blur filtering with Laplacian variance (32,648 images remaining), face detection and cropping with Haar Cascade, data augmentation, and class imbalance handling with WeightedRandomSampler.
- Fine-tuned an ImageNet-pretrained ResNet50 in two stages, reaching 85.76% validation accuracy and macro-F1 of 0.85 (about 18 points higher than training the final layer only).

SKILLS

AI and Data: Large Language Models (LLM), Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Prompt Engineering, Machine Learning, Predictive Modeling, Deep Learning, Computer Vision, Sentiment Analysis, Model Evaluation (RAGAS), Feature Selection (SHAP), Class Imbalance Handling, Data Preprocessing, Data Visualization.

Programming and Databases: Python, SQL, MySQL, JavaScript, PHP.

Framework and Tools: PyTorch, Scikit-learn, XGBoost, Hugging Face Transformers, LangChain, FAISS, Pandas, Seaborn, OpenCV, NLTK, Gradio, Laravel, React, Git, Jupyter Notebook, Google Workspace, Microsoft Office.

Supporting: Team leadership and coordination, stakeholder communication, public speaking (MC, moderator, speaker), technical presentation, critical and analytical thinking, problem solving, documentation.

Languages: Indonesian; English (TOEFL ITP: 550).

CERTIFICATIONS AND COURSES

Associate Data Scientist (CADS) - BNSP *2026*
Gemini Certified University Student - Google *2026*
Artificial Intelligence & Automation Bootcamp - MySkill *2026*
AI for Work & Career Readiness With Google AI Products - Hacktiv8 *2026*
Data Analyst Fundamental - MySkill *2026*
Google AI Essentials - Google *2025*
Software Engineer Fundamental Course - RevoU *2024*